

Airbnb Barcelona Price Prediction

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The Problem

How much should an Airbnb listing in Barcelona cost per night?

Hosts

Often **misprice** their listings — too high or too low

Guests

Overpay or miss good deals without a reference point

Market

Dynamic pricing is **complex** and hard to estimate manually

The Dataset

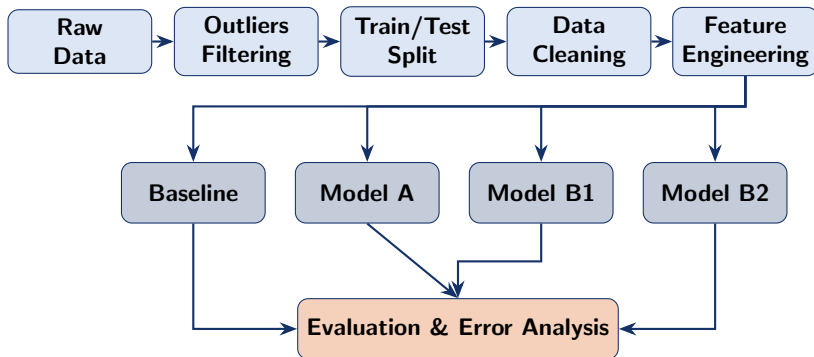
Property	Value
City	Barcelona, Spain
Total listings	19,410
Features	79
Target variable	Nightly price (€)
Listings with price	15,276
Final training set	9,426
Final test set	3,056

The Dataset

Real-world challenges in this dataset

- Heavily **skewed** price distribution
- **Structural** missing values in review columns (~23% of listings)
- Mixed feature types: numeric, categorical, text, geographic

Pipeline Overview



Target Variable: Price

Raw format

- Stored as string: "\$100.00"
- Strip \$ and , $\rightarrow$ cast to float
- Remove 4,134 rows with missing price

Log transform

Price is heavily right-skewed, so:

$$\tilde{y} = \log(p + 1)$$

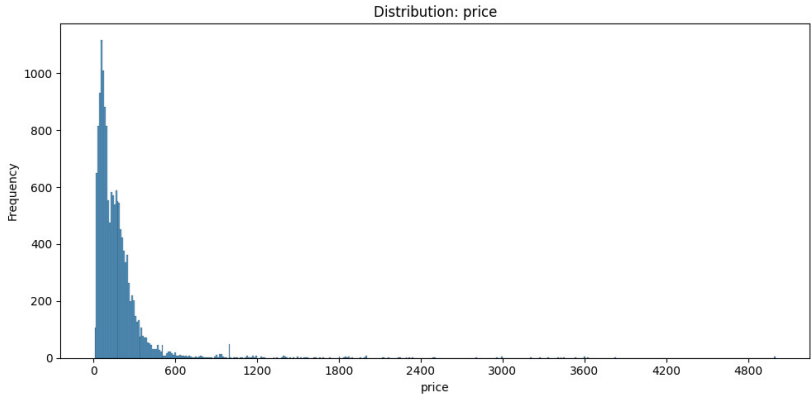
Train the model in log space, then recover:

$$\hat{p} = e^{\hat{y}} - 1$$

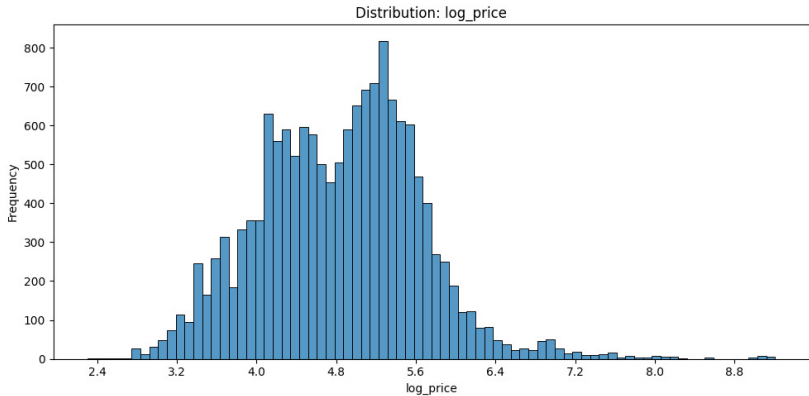
Why log1p?

- Compresses the scale
- Reduces influence of luxury outliers (€9k–€10k)
- Makes the distribution closer to Gaussian
- Improves linear separability

Target Variable: Price



Target Variable: Price



Outlier Filtering

Problem detected from initial training

- $RMSE \gg MAE$
- $R^2(\text{€}) \ll R^2(\text{log})$
- Suggests a small number of listings with **extreme prices** dominating the error

Filtering criterion

Remove listings where:

$$\text{price} > \text{€}2,000$$

Feature Selection & Missing Values

Feature selection

- 79 raw columns → **36 retained**
- Dropped columns:
 - Listing URLs, IDs
 - Free-text descriptions
 - Columns with >50% missing
(e.g., `calendar_updated`: 100% null)

Feature Selection & Missing Values

Missing value strategy

Column group		Strategy
Non-review cols	(<6% missing)	Median / mode imputation
Review cols	(~23% missing)	Separate model (see later)

All imputation statistics computed on **training set only** to prevent data leakage.

Feature Engineering — Host & Property

Host features

Feature	Transform
host_since	Days since scrape
response_time	Ordinal (1→36)
response_rate	Strip %, ÷100
acceptance_rate	Strip %, ÷100
superhost, verified...	0/1 flag

Feature Engineering — Host & Property

Room type

- One-hot → 4 binary columns:
`entire_home`, `private_room`,
`shared_room`, `hotel_room`

Bathrooms

- Parse `bathrooms_text`:
 - `baths`: float (0.5 for half-bath)
 - `shared_bath`: binary flag

Property & Neighbourhood

- **Target encoding** with smoothing
- Prevents overfitting on rare categories

Feature Engineering — Amenities

Processing amenities required a **3-step pipeline**:

Step 1 — Normalize

Group semantically similar names:

e.g., all Wi-Fi variants → `wifi`,
multiple TV sizes → `tv`,
streaming services →
`streaming_service`

Step 2 — Binarize

Multi-label binarization:
each unique amenity → one
binary column
(115 columns total)

Step 3 — Correlation select

Keep top **12** amenities by
`|corr(amenity, log(price))|`

Feature Engineering — Geographic Distance

Approach

- Replace raw latitude/longitude with **Haversine distances** to 5 landmarks
- More interpretable than coordinates
- Captures spatial price gradient

Haversine formula

$$d = 2R \arcsin \left(\sqrt{\sin^2 \frac{\Delta\phi}{2} + \cos \phi_1 \cos \phi_2 \sin^2 \frac{\Delta\lambda}{2}} \right)$$

Feature Engineering — Geographic Distance

Landmark features

Feature	Landmark
dist_sagrada	Sagrada Família
dist_ramblas	La Rambla
dist_barceloneta	Barceloneta Beach
dist_campnou	Camp Nou
dist_catalunya	Pl. Catalunya

The Structural Missingness Problem

Observation: 8 columns are missing *jointly* on the same $\sim 23\%$ of listings

Column	Missing
<code>review_scores_rating</code>	22.86%
<code>review_scores_accuracy</code>	22.87%
<code>review_scores_cleanliness</code>	22.87%
<code>review_scores_checkin</code>	22.87%
<code>review_scores_communication</code>	22.86%
<code>review_scores_location</code>	22.86%
<code>review_scores_value</code>	22.86%
<code>reviews_per_month</code>	22.86%

The Structural Missingness Problem

Root cause

These listings have **never received a guest review**.

Why not just impute the median?

- This is **not** random missingness
- Median imputation introduces **fake signal**
- The *absence* of reviews is itself informative

Our solution

Train **separate specialist models** for reviewed vs. non-reviewed listings

Multi-Model Architecture

Model	Trained on	Review columns	Role
Baseline (bm)	All training data	Keep + fill 0 + has_review flag	Single model for everything (comparison anchor)
Model A	Only samples with reviews	Keep (use as features)	Expert for reviewed listings
Model B1	Only samples without reviews	Drop 8 cols	No-review expert (less data, exact segment)
Model B2	All training data	Drop 8 cols	No-review expert (more data, shift risk)

Model Training - Linear/ Polynomials Regression

Implementation

- Framework: **PyTorch**
- Architecture: Linear Regression (proof of concept baseline)
- Loss: Mean Squared Error (MSE)
- Optimizer: Adam
- Input normalization: (mean/std from training set only)

Data split

Training set	9,426 listings
Test set	3,056 listings

Other Models:

- Random Forest
- Gradient Boosting
- LightGBM (recommended by Claude)

Results metrics - Linear Regression

===== LinReg bm =====

RMSE: €117.87

MAE: €56.35

Median AE: €28.37

R^2 : 0.5232

R^2 (log-scale): 0.6787

The dataset B1 is
for future works

===== LinRegA =====

RMSE: €118.55

MAE: €55.22

Median AE: €29.20

R^2 : 0.5177

R^2 (log-scale): 0.6849

===== LinReg B2 =====

RMSE: €119.60

MAE: €57.62

Median AE: €32.10

R^2 : 0.5091

R^2 (log-scale): 0.6639

Results metrics - LightGBM

===== LGBM bm =====

RMSE: €95.88

MAE: €40.43

Median AE: €18.88

R²: 0.6845

R² (log-scale): 0.8131

===== LGBM B1 =====

RMSE: €117.79

MAE: €59.61

Median AE: €27.84

R²: 0.5239

R² (log-scale): 0.6805

===== LGBM A =====

RMSE: €103.33

MAE: €43.08

Median AE: €20.17

R²: 0.6336

R² (log-scale): 0.7759

===== LGBM B2 =====

RMSE: €94.80

MAE: €39.35

Median AE: €18.60

R²: 0.6916

R² (log-scale): 0.8153

Limitations & Future Work

Current limitations

- Luxury listings ($\text{€} > 1,000$) poorly predicted
- Single data scrape: No seasonal/temporal dynamics
- Did not use name, description and image data yet.
- Listing with high accomodate create "reasonable" outliers.

Limitations & Future Work

Future directions

- Other models: MLP,...
- Two-stage model: classify luxury/standard → regress
- NLP features from listing descriptions
- Change the target to `price_per_person`